



Math, Science, and Engineering
Invitational

2026

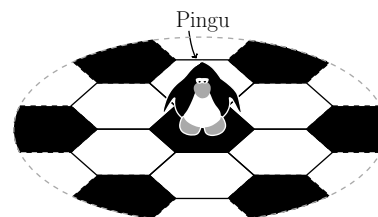
Problems

Team Round - 30 Points:

1. How many three-digit positive integers have the property that their tens digit is two times their units digit?

2. Pingu is tiling his floor with black and white regular hexagonal tiles.

Any black tile must have exactly six white tiles surrounding it. If Pingu's floor needs exactly 39 tiles, what is the maximum amount of black tiles he can use?



3. Let a, b be relatively prime positive integers such that

$$\frac{6}{1 + \frac{5}{2 + \frac{4}{3}}} = \frac{a}{b}$$

If a and b are the legs of a right triangle, compute the perimeter of the triangle.

4. Let $\underline{ONE}$ and $\underline{TWO}$ be three digit integers with digits O, N, E and T, W, O respectively such that

$$\begin{array}{r} \text{O} \text{ N} \text{ E} \\ + \text{O} \text{ N} \text{ E} \\ \hline \text{T} \text{ W} \text{ O} \end{array}$$

where T, O, E are all even, and $\underline{TWO}$ is divisible by 3. Compute one possible value of $T + W + O$.

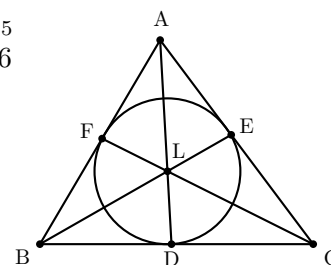
5. A number is *patriotic* if we can delete some number of its digits without changing the order of the digits and be left with the number 1776 (for example, 2133776 is *patriotic* because the digits 2, 3, and 3 can be deleted to yield 1776). How many *patriotic* numbers are less than 100,000?

6. Mariam has x shirts, y skirts, and z hats. By picking one of each she can create n unique outfits. If Vio steals 3 of her shirts and 1 hat, Mariam can only make $\frac{n}{3}$ outfits. If Jamie borrows 2 skirts and 2 hats she can only make $\frac{n}{5}$ outfits. If she buys 4 shirts and 1 skirt she can now make $2n$ outfits. What is n ?

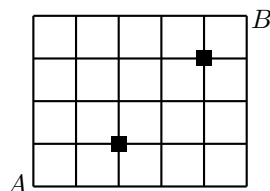
7. Rectangle $MSEI$ is reflected over its diagonal SI , sending M to A and E to C , forming a new rectangle $ASCI$. Let X be the intersection of AC and MS . If $CX = 1$ and $\angle ASE = 30^\circ$, find the area of $MSEI$.

8. Consider the integers x_1, x_2, x_3, x_4, x_5 . The sequences x_1, x_2, x_3 and x_3, x_4, x_5 are arithmetic sequences and x_2, x_3, x_4 forms a geometric sequence. If $x_1 = 6$ and $x_5 = 20$, find x_3 .

9. Let ABC be a triangle where $AB = 13$, $AC = 14$ and $BC = 15$. Let D, E , and F be the tangent points of the incircle of ABC to BC, AC , and AB , respectively. Let L be the intersection of AD, BE , and CF . Find AL/AD .



10. Suppose a traveling salesman is standing at $A : (0, 0)$ and is trying to walk to $B : (5, 4)$. The salesman can move either one unit up or one unit right at a time. Given that on $(2, 1)$ and $(4, 3)$, there exists a one-time use teleporter that takes you to the other location (that is, if you stand on the teleporter at $(2, 1)$, you get sent to $(4, 3)$, and if you stand on it at $(4, 3)$, you get sent to $(2, 1)$), compute the number of ways the salesman can go from A to B . (If he has already used the teleporter, he is able to walk over it.)



Happy New Year!

The Power Round lasts 60 minutes with an additional 10 minute reading period. During this time, teams will work to perform calculations and write solutions based on the given background information. While working on a problem, you may use any confirmed theorems or assertions in the problem statements of previous problems, even if your team has not proved those theorems or solved those problems. For the sake of clarity:

- Any question that uses the words “compute” or “find” requires an answer. Work shown on the answer sheet is recommended to support your result.
- Any question that uses the word “justify” requires a brief explanation.
- Any question that uses the words “show” or “prove” requires rigorous justification of the result.

To receive full credit, the presentation must be legible, orderly, clear, and concise. The pages submitted for credit should be in consecutive question order. Please write on one side of the answer paper only. Put the room number (not the team name) on all pages submitted. Do not identify the team in any other way.

Background Information - 10 Minutes:

Let $S(n)$ denote the sum of the digits of n , $S_2(n)$ denote the sum of the squares of the digits of n , and $S_3(n)$ denote the sum of the cubes of the digits of n .

Define a **happy number** k as any positive integer that, when iterated over $S_2(n)$, eventually reaches 1. For instance, 13 is a happy number because $1^2 + 3^2 = 10$, and $1^2 + 0^2 = 1$. Generalizing, a positive integer n is happy if and only there exist some integer $m \geq 0$ such that $S_2^m(n) = 1$, where m is the number of iterations. For the example above, we would say $S_2(13) = 10$, and $S_2^2(13) = S_2(S_2(13)) = 1$.

The **height** of a happy number n is the least integer $m \geq 0$ such that $S_2^m(n) = 1$. A few examples, 1 is a happy number of height 0, 10 is a happy number of height 1, and 13 is a happy number of height 2.

Given integer $b \geq 1$, in a base b system, our highest digit is $b - 1$. Note that regularly, we use a base-10 system as our highest digit is 9, but for example, in base 6, our highest digit is 5. To represent numbers higher, we perform the same trick we do in base 10, which is using place value to denote the exponent. For example, in base-10, $125 = \underline{1} \cdot 10^2 + \underline{2} \cdot 10^1 + \underline{5} \cdot 10^0$, but in base 6, the number 125 is $\underline{3} \cdot 6^2 + \underline{2} \cdot 6^1 + \underline{5} \cdot 6^0$ so we write it as 325_6 .

A **b -happy number** is defined as a positive integer in a given base b that eventually reaches 1 when iterated over $S_2(n_b)$, where n_b is the integer n in base b . For example, 347 is a 6-happy number:

$$S_2(347) = S_2(1335_6) \rightarrow 1^2 + 3^2 + 3^2 + 5^2 = 44$$

$$S_2(44) = S_2(112_6) \rightarrow 1^2 + 1^2 + 2^2 = 6$$

$$S_2(6) = S_2(10_6) \rightarrow 1^2 + 0^2 = 1.$$

We may generalize the concept of a happy number and say that a positive integer j is a **cubic happy number** if when iterated over $S_3(n)$, eventually it will reach 1. Formalizing, a positive integer n is cubic happy if and only if there exist some integer $m \geq 0$ such that $S_3^m(n) = 1$, where m is the number of iterations.

Define the **cubic height** of a cubic happy number as the least integer $m \geq 0$ such that $S_3^m(n) = 1$.

Power Questions - 30 Points:

1. [4] Compute if the following numbers are happy.
 - (a) $n = 2026$
 - (b) $n = 67$
 - (c) $n = 68$
 - (d) $n = 167$
 2. [2] Given $n = 356$ is a happy number, compute its height.
 3. [2] Compute the smallest positive integer n with height 3.
 4. [4] Compute if the following numbers are cubic happy. If n is cubic happy, provide the cubic height of n . Show work to receive full credit.
 - (a) $n = 153$
 - (b) $n = 370$
 - (c) $n = 371$
 - (d) $n = 407$
 5. [3] Compute the greatest positive integer n such that $S_3(n) > n$.
 6. Define a **happy base** as any base b , where b is a positive integer, such that all positive integers in base b are happy.
 - (a) [1] Convert 45 into base 2.
 - (b) [1] Show that 45 is happy in base 2.
 - (c) [3] Show that base 2 is a happy base.
 7. Given integers a, b , and c , where $c \neq 0$, $a \equiv b \pmod{c}$ if c divides $a - b$, or more informally, if a leaves a remainder of b when divided by c . Given n is a positive integer:
 - (a) [3] Prove that for all integers $m \geq 0$, $S_3^m(n) \equiv n \pmod{3}$.
 - (b) [3] Prove that a positive integer n is cubic happy only if $n \equiv 1 \pmod{3}$.
 - (c) [1] Prove that a positive integer $n \equiv 1 \pmod{3}$ does not imply that n is cubic happy.
 8. [3] Compute the smallest positive integer n with cubic height 3.
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Guts Problems - All Sets:

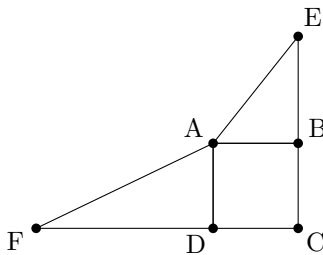
1. Compute the last digit of $9^6 + 9^7$.
 2. The average weight of the 8 girls in a class is 125. The average weight of the 20 boys in the class is 160. What is the average weight of all the students in the class?
 3. Dashiell writes 10 problems per week. Jayson rejects 90% of his problems. How many weeks will it take him to write 10 acceptable problems?
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4. The first three terms (in order) of an arithmetic progression are $x + 1$, $2x$, and $2x + 3$. Compute all possible values of x .
 5. Jayson flips 8 coins. What is the probability that he flips an odd number of tails?
 6. You have been paired up with another team at MSEI. Both of you are greedily looking at a treasure trove of 2 points. You both have the option to either split or steal the trove. If you both split, you both receive 1 point. If one of you steals and the other one splits, the team that stole gets 2 points, and if you both steal, nobody gets any points. Write 1 to steal or 0 to split as your answer.
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7. Mr. Vu pulls two socks at random from a drawer which contains 6 identical white socks and 7 identical black socks. Compute the probability the socks match.
 8. The minute hand of a clock travels a total of $x\pi$ radians between 11 am and 1 : 15 pm that same afternoon. Compute x .
 9. Given a 3-4-5 right triangle, compute the length of the altitude to the hypotenuse.
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10. The integer 1234321 is a perfect square. Compute $\sqrt{1234321}$.
 11. Triangle ABC has right angle B and AB is 7 and BC is 23. Let point D be the point where the median from angle B is drawn to AC . What is the ratio of the area of BCD to ABD ?
 12. Let P be the number you submit, and Q be the number of (uppercase) M's, S's, E's, and I's we printed for today's competition. You will receive $\frac{1}{\sqrt[5]{|P-Q|}}$ points. In the case $P = Q$, you will receive 2 points.
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13. Compute all integer solutions to $(x + 5)(x + 7) = 89999$.
14. In the diagram below, $ABCD$ is a square. Given that $\triangle ABE$ has area 14, $\triangle ADF$ has area 35, $\triangle CEF$ has area $\frac{187}{2}$, and $EB \cdot DF = 40$, compute the length of segment EF .



15. Let $f(x) = x^2 + nx + m$ be a quadratic with roots $\sin \theta + 1$ and $\cos \theta + 1$. What is the maximum value of $m + n$?

16. Let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x . Compute the smallest real $x > 0$ such that

$$x \cdot \lfloor x \rfloor = x^2 - \frac{1}{2}x - 3.$$

17. In $\triangle ABC$ with $AB = 3$, $BC = 4$, and $CA = 5$, point X is chosen on BC such the the distance from X to line AB equals the distance from X to line AC . If the foot of the perpendicular from X to AC is Y , compute the area of $\triangle XCY$.
18. The company wants you to approximate the value of $\sqrt{2}$. Submit a decimal n and you will receive $\log x$ points, rounded to the nearest ten thousandths, where x is largest the number of digits in a row that match $\sqrt{2}$'s decimal expansion, starting after the decimal point. For example, if you were trying to approximate $\pi \approx 3.14159265$ and you submitted 108.14157263, you would receive $\log 4 = 0.6021$ points.

19. Define the sequence $a_1, a_2, \dots$ where $a_1 = 1$ and a_n is the remainder of $3a_{n-1}$ when divided by 17. Find the average value across all the values of the sequence.
20. For prime p and positive integer k , $k! + p = p^k = n$. Compute n .
21. Rumbles the cat jumps down a row of 90 pillows, going from pillow 4 to 8 to 12 and continuing to go 4 pillows forward with each jump. Once his last jump brings him beyond the 90th pillow, Rumbles returns back by jumping 3 pillows at a time, starting with the 3rd to last pillow and then the 6th to last pillow and so on. How many pillows does he jump on an even number of times?

22. Suppose real numbers x and y exist that satisfy the equation

$$\begin{aligned} xy &= 64 \\ x^{\log_2 y} &= 256 \end{aligned}$$

Compute $x + y$.

23. Compute the number of prime numbers that are in the prime factorization of $2027!$, but not $2026!$.
24. Submit a composite integer from 6 to 100, inclusive. For every other submission that is relatively prime to yours, you will receive $\frac{1}{5}$ points. If the integer you submit is prime or out of the range, you will lose 1 point.

25. Let a_n be a geometric sequence where $a_1 = 2$ and $a_3 = -18$. Compute the maximum value of a_8 .
26. For positive integers A and B , the numbers A , B^2 , A^2 , and $A^2 + B^2 - 8$ form an arithmetic sequence. Compute the common difference of this sequence.
27. Suppose x is a positive real number. The minimum value of

$$2026x^2 + \frac{1}{1013x}$$

can be written in the form $\frac{m}{\sqrt[3]{n}}$ for integers m and n where m has no perfect cube factors. What is $m + n$?

28. Gatsby has a deck of 40 cards, arranged in order as the ace through 10 of clubs, hearts, spades, and diamonds. Gatsby makes some number of cuts, with each cut removing n cards from the top of the deck and putting them on the bottom of the deck in the same order. He then alternates dealing himself and Daisy 1 card, dealing each player 20 cards. If the aces are worth 1 point and each other card is worth its number of points, compute all possible remainders when the sum of the point values of all of Gatsby's cards is divided by 67.
29. If

$$\sin \theta \cdot \cos \theta \cdot \cos 2\theta \cdot \cos 4\theta \cdot \cos 8\theta \cdot \dots \cdot \cos 128\theta = \frac{2\theta}{\pi},$$

compute all possible θ .

30. List as many primes as you can. For each prime p you list, you will receive $\frac{1}{p}$ points. However, if you list even a single non-prime number, you will receive no points.

Note: please write legibly and unambiguously. If a number is not easy to read or could be interpreted as having multiple values, it will be ignored.

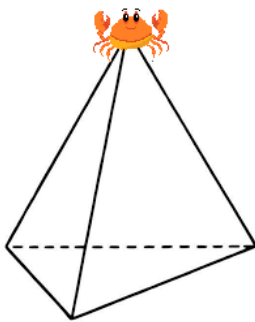
Individual Problems - All Sets:

1. Given k is positive, if $k\%$ of k is 2025, compute k .
2. Arvin is trying to copy his favorite 90s bands, such as Nirvana. Compute the number of ways to rearrange the letters in **ANARVINNIRVANA** such that the rearranged string is a palindrome (a palindrome is a word that is the same written forwards and backwards).

3. 100 people stand in a line, each of which is a knight or a knave. Knights always tell the truth and knaves always lie. When asked if there are an even number of knights behind them in the line, each person answers yes. How many knaves are in the line?
4. Given $3x^2 + 2x + 7 = 0$ has roots r and s , find $\frac{1}{r} + \frac{1}{s}$.

5. Call a quadratic polynomial “guessable” if both of its roots are integers. Find the sum of the linear coefficients of all guessable polynomials with coefficients between -10 and 10 , inclusive.
6. What is the greatest number of unique digits a number N can have if the numbers formed by any two adjacent digits of N is divisible by 7? (For example, 214 would be a valid number because 21 and 14 are both divisible by 7).

7. Let $S(n)$ denote the sum of the digits of n . Compute the leftmost 3 digits of the smallest integer n such that $S(S(n)) = 21$.
8. Sandy the Crab is standing on a vertex of a tetrahedron. Each minute Sandy moves to a random different vertex. Compute the probability that Sandy has visited each vertex of the tetrahedron at least once after 4 minutes have passed (the starting vertex is considered to have been visited at the start).

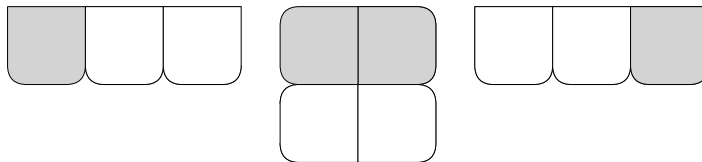


9. The sum of 2026 distinct prime numbers divides their product. Let n be the smallest of these prime numbers. Compute all possible values of n .
10. Maya and Anna, who are perfect logicians, decide to play a game of *triangle*. To play a game of *triangle*, they first construct the board from an equilateral triangle with area 1, splitting it into 4 similar triangles by connecting the midpoints of the sides. They do the same on the middle of the 4 triangles, repeating this process an infinite number of times.

To play *triangle*, Anna and Maya take turns claiming triangles on the board with Anna starting. After the entire board is claimed (in infinite turns), each player's score is the total area of the triangles they have claimed. How much will Anna win by, assuming both Maya and Anna play optimally?

Tiebreakers:

1. A section of a R68 train car has 10 seats. The 4 seats shaded on the diagram are more desirable and will be chosen over the unshaded seats if unoccupied. Given 3 people get on at 7th avenue station and 3 people get on at Atlantic Avenue-Barclays Center, how many possible seating arrangements are possible as the train enters DeKalb Avenue (the next station)?



2. How many 4-digit integers $MSEI$ satisfy

$$MSE > SEI$$

$$IES > ESM$$

Where M , S , E , and I are digits and none of the digits are 0?

3. Consider the sequence of spheres $\omega_1, \omega_2, \omega_3, \dots$ where each sphere has half the radius of the last and ω_1 has radius 1. If these spheres are arranged in order such that sphere ω_n is externally tangent to sphere ω_{n+1} and ω_{n-1} where the centers of all the spheres lie on a line, compute the ratio between the total volume of the spheres and the volume of the smallest cone that contains all the spheres.